TECHNICAL DOCUMENTATION
EMPLOYEE MANAGEMENT SYSTEM (EMS)

1. INTRODUCTION

The Employee Management System (EMS) is designed to organize and manage employee information, departments, job positions, salaries, and project assignments. The database uses related tables to keep records organized, reduce unnecessary duplication, and make employee information easier to retrieve and maintain.

2. OBJECTIVES

- To store employee personal and employment information.
- To organize department and position details.
- To record employee salary information.
- To connect employees with the projects they work on.
- To maintain accurate, consistent, and organized records.
- To make searching and managing employee information easier.

3. DATABASE OVERVIEW

Database name: employee_management

The database contains five related tables:

Table Name	Description	Purpose
employees	Stores employee personal and employment information.	Employee records
departments	Stores department names and locations.	Department records
positions	Stores employee position titles.	Position records
projects	Stores project names and start dates.	Project records

employee_projects	Links employees to projects and stores their project roles.	Employee-project assignments

4. ENTITY RELATIONSHIP DIAGRAM (ERD)

The ERD shows how the five tables are connected. A department can have many employees, and a position can be assigned to many employees. Employees and projects have a many-to-many relationship, managed through the employee_projects junction table.

5. DATABASE NORMALIZATION

Normalization organizes data into related tables to reduce duplicate information and improve data consistency.

First Normal Form (1NF)

The database follows 1NF because each field stores one value only, and each record can be identified by a key. For example, first_name and last_name are stored in separate fields instead of one combined name field.

Second Normal Form (2NF)

The database follows 2NF because non-key fields depend on the whole primary key. In employee_projects, employee_id and project_id form a composite primary key, while role_in_project describes that employee's assignment to that specific project.

Third Normal Form (3NF)

The database follows 3NF by storing department details in departments, position details in positions, project details in projects, and employee-project assignments in employee_projects. This avoids repeating department, position, and project details in employee records and makes updates easier.

6. DATABASE RELATIONSHIPS

Relationship	Type	Explanation
Departments → Employees	One-to-many (1:M)	One department can have many employees; each employee belongs to one department.
Positions → Employees	One-to-many (1:M)	One position can be assigned to many employees; each employee has one position.
Employees → Employee_Projects	One-to-many (1:M)	One employee can have multiple project assignments.
Projects → Employee_Projects	One-to-many (1:M)	One project can have multiple employee assignments.

7. DATA DICTIONARY

A data dictionary is a document that describes the tables, fields, data types, keys, and purposes of the data stored in a database. It helps developers understand how the database is organized and how the tables are connected.

How We Created the Data Dictionary

We created the data dictionary by identifying the tables needed for the Employee Management System, listing the fields in each table, assigning the appropriate data types, and defining the keys such as Primary Key (PK), Foreign Key (FK), and Unique. We also added descriptions to explain the purpose of each field.

The Tables

- **Employees** – Stores employee information, including names, contact details, salary, status, department, and position.

- **Departments** – Stores department names and their locations.

- **Projects** – Stores project information, including project names and start dates.
- **Employee_Projects** – Connects employees to projects and records their roles in each project.
- **Positions** – Stores employee position titles and their unique IDs.

Keys and Data Types

- **PK (Primary Key)** – Uniquely identifies each record in a table.

- **FK (Foreign Key)** – Connects a table to another table.
- **UNIQUE** – Prevents duplicate values in a field.
- **INT** – Stores whole numbers.
- **VARCHAR** – Stores text with a specified maximum length.
- **DATE** – Stores dates.
- **DECIMAL(10,2)** – Stores numbers with two decimal places, such as salaries.
- **ENUM** – Allows values from a predefined list.
- **CHECK** – Ensures that a value meets a specified condition.

8. CONCLUSION

The EMS database provides an organized structure for managing employees, departments, positions, and projects. Its relationships and normalization help reduce duplicate data and maintain consistency. The ERD and data dictionary serve as guides for developing, maintaining, and understanding the system.